

ECEN-601: Homework 2

Due on September 19, 2025 at 5:00 pm

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Problem 1

Negate the following statement: For every real number $\varepsilon > 0$ there exists a positive integer k such that for all positive integers n , it is the case that $|a_n - k^2| < \varepsilon$.

Solution

There exists some $\varepsilon > 0$ such that for each positive integer k , there exists some positive integer n where $|a_n - k^2| \geq \varepsilon$.

Problem 2

Let $f : X \rightarrow Y$, $A, B \subseteq X$, and $C, D \subseteq Y$. Show that f^{-1} preserves inclusions, unions, and intersections.

[Hint: Recall that $f(A) \triangleq \{f(x) \mid x \in A\}$ and $f^{-1}(C) \triangleq \{x \in X \mid f(x) \in C\}$.]

- (a) Show that $C \subseteq D \implies f^{-1}(C) \subseteq f^{-1}(D)$.
- (b) Show that $f^{-1}(C \cup D) = f^{-1}(C) \cup f^{-1}(D)$.
- (c) Show that $f^{-1}(C \cap D) = f^{-1}(C) \cap f^{-1}(D)$.

Show that f preserves inclusions and unions, but not intersections.

- (d) Show that $A \subseteq B \implies f(A) \subseteq f(B)$.
- (e) Show that $f(A \cup B) = f(A) \cup f(B)$.
- (f) Show that $f(A \cap B) \subseteq f(A) \cap f(B)$. Give an example where equality does not occur.

Solution

Part (a): Say we have $x \in f^{-1}(C)$, then $f(x) \in C$, which implies $x \in C$. If we assume $C \subseteq D$, then $x \in D$, which means $f(x) \in D$, and if that is true, then $x \in f^{-1}(D)$. Thus, $f^{-1}(C) \subseteq f^{-1}(D)$.

Part (b): Say we have $x \in f^{-1}(C \cup D)$, then $f(x) \in (C \cup D) \leftrightarrow f(x) \in C \cup f(x) \in D$. By the definition of the preimage, this means $x \in f^{-1}(C) \cup f^{-1}(D)$. Thus, $f^{-1}(C \cup D) = f^{-1}(C) \cup f^{-1}(D)$.

Part (c): Say we have $x \in f^{-1}(C \cap D)$, then $f(x) \in (C \cap D) \leftrightarrow f(x) \in C \cap f(x) \in D$. By the definition of the preimage, this means $x \in f^{-1}(C) \cap f^{-1}(D)$. Thus, $f^{-1}(C \cap D) = f^{-1}(C) \cap f^{-1}(D)$.

Part (d): Say we have $x \in A$, then $f(x) \in f(A)$. If we assume $A \subseteq B$, then $x \in B$, which means $f(x) \in f(B)$. Thus, $A \subseteq B \implies f(A) \subseteq f(B)$.

Part (e): Say we have $x \in f(A \cup B)$, then $f(x) \in f(A \cup B)$, which means $f(x) \in f(A) \cup f(x) \in f(B)$. Thus, $f(A \cup B) = f(A) \cup f(B)$.

Part (d): Say we have $x \in f(A \cap B)$, then $x \in A \cap B$. This means $f(x) \in f(A)$ and $f(x) \in f(B)$. Since $f(x)$ is in both sets, $f(A \cap B) \subseteq f(A) \cap f(B)$.

Problem 3

Let $\underline{x} = (x_1, \dots, x_n)$, $\underline{y} = (y_1, \dots, y_n) \in \mathbb{R}^n$ and consider the function ρ given by

$$\rho(\underline{x}, \underline{y}) = \max |x_1 - y_1|, \dots, |x_n - y_n|.$$

Show that ρ is a metric.

Solution

In order for ρ to be a metric, it must satisfy the following properties

1. $\rho(x, y) \geq 0 \quad \forall x, y \in X : \text{equality holds if and only if } x = y$
2. $\rho(x, y) = \rho(y, x) \quad \forall x, y \in X$
3. $\rho(x, y) + \rho(y, z) \geq \rho(x, z) \quad \forall x, y, z \in X$

As stated in the problem, we are working over the real numbers.

Starting from the top:

$$\rho(\underline{x}, \underline{y}) = \max |x_1 - y_1|, \dots, |x_n - y_n| \geq 0$$

This inequality holds because we are taking the max of absolute values, so we will never get a negative number. Additionally, over the reals, we only get 0 when $x_i = y_i$. Thus, ρ satisfies non-negativity.

Next, we check if $\rho(\underline{x}, \underline{y}) = \rho(\underline{y}, \underline{x})$:

$$\rho(\underline{x}, \underline{y}) = \max |x_1 - y_1|, \dots, |x_n - y_n| = \max |y_1 - x_1|, \dots, |y_n - x_n| = \rho(\underline{y}, \underline{x})$$

We know this is true since $|a - b| = |b - a|$ for $a, b \in \mathbb{R}$. Thus, ρ satisfies symmetry.

Finally, the triangle inequality:

$$\begin{aligned} \rho(\underline{x}, \underline{y}) + \rho(\underline{y}, \underline{z}) &\geq \rho(\underline{x}, \underline{z}) \\ \max |x_1 - y_1|, \dots, |x_n - y_n| + \max |y_1 - z_1|, \dots, |y_n - z_n| &\geq \max |x_1 - z_1|, \dots, |x_n - z_n| \end{aligned}$$

Holds since $|x_k - z_k| = |(x_k - y_k) + (y_k - z_k)|$ and each term inside the absolute value is upper bounded by the maximum value.

ρ satisfies all three properties, so it is a metric.

Problem 4

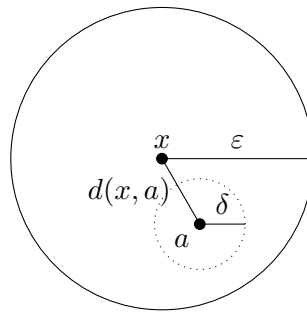
Suppose $a \in B_d(x, \varepsilon)$ with $\varepsilon > 0$. Find an explicit $\delta > 0$ such that the open ball $B_d(a, \delta)$ centered at a is contained in $B_d(x, \varepsilon)$.

Solution

We have a set $B_d(x, \varepsilon) = \{y \in X \mid d(x, y) < \varepsilon\}$ in which we find a . This means that $d(x, a) < \varepsilon$.

We need some δ such that $\forall y \in B_d(a, \delta), y \in B_d(x, \varepsilon)$.

I think this problem is very easy to solve visually:



As we can see above, δ is simply less than the difference between ε and $d(x, a)$. Thus, for any δ such that $\delta < \varepsilon - d(x, a)$, $B_d(a, \delta) \subset B_d(x, \varepsilon)$.

Problem 5

This problem outlines a proof that the Euclidean distance d on $\mathbb{R}^n$ is a metric, as follows: If $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ and $c \in \mathbb{R}$, define

$$\begin{aligned}\mathbf{x} + \mathbf{y} &= (x_1 + y_1, \dots, x_n + y_n) & c\mathbf{x} &= (cx_1, \dots, cx_n), \\ \mathbf{x} \cdot \mathbf{y} &= x_1y_1 + \dots + x_ny_n & \|\mathbf{x}\| &= (\mathbf{x} \cdot \mathbf{x})^{1/2}.\end{aligned}$$

- (a) Show that $\mathbf{x} \cdot (\mathbf{y} + \mathbf{z}) = (\mathbf{x} \cdot \mathbf{y}) + (\mathbf{x} \cdot \mathbf{z})$.
- (b) Show that $|\mathbf{x} \cdot \mathbf{y}| \leq \|\mathbf{x}\| \|\mathbf{y}\|$. [Hint: if $\mathbf{x}, \mathbf{y} \neq 0$, let $a = 1/\|\mathbf{x}\|$ and $b = 1/\|\mathbf{y}\|$, and use the fact that $\|a\mathbf{x} \pm b\mathbf{y}\| \geq 0$.]
- (c) Show that $\|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|$. [Hint: Compute $(\mathbf{x} + \mathbf{y}) \cdot (\mathbf{x} + \mathbf{y})$ and apply previous result.]
- (d) Verify that the Euclidean distance $d(\mathbf{x}, \mathbf{y})$ is a metric.

Solution

Part (a): Very simply,

$$\begin{aligned}\mathbf{x} \cdot (\mathbf{y} + \mathbf{z}) &= x_1(y_1 + z_1) + \dots + x_n(y_n + z_n) \\ &= x_1y_1 + x_1z_1 + \dots + x_ny_n + x_nz_n \\ &= (x_1y_1 + \dots + x_ny_n) + (x_1z_1 + \dots + x_nz_n) \\ &= (\mathbf{x} \cdot \mathbf{y}) + (\mathbf{x} \cdot \mathbf{z})\end{aligned}$$

Part (b): Let $a = 1/\|\mathbf{x}\|$ and $b = 1/\|\mathbf{y}\|$. Then,

$$\begin{aligned}0 \leq \|a\mathbf{x} + b\mathbf{y}\|^2 &= (a\mathbf{x} + b\mathbf{y}) \cdot (a\mathbf{x} + b\mathbf{y}) \\ &= a^2(\mathbf{x} \cdot \mathbf{x}) + 2ab(\mathbf{x} \cdot \mathbf{y}) + b^2(\mathbf{y} \cdot \mathbf{y}) \\ &= a^2\|\mathbf{x}\|^2 + 2ab(\mathbf{x} \cdot \mathbf{y}) + b^2\|\mathbf{y}\|^2 \\ &= \frac{\|\mathbf{x}\|^2}{\|\mathbf{x}\|^2} + \frac{2(\mathbf{x} \cdot \mathbf{y})}{\|\mathbf{x}\|\|\mathbf{y}\|} + \frac{\|\mathbf{y}\|^2}{\|\mathbf{y}\|^2} \\ &= 2 + \frac{2(\mathbf{x} \cdot \mathbf{y})}{\|\mathbf{x}\|\|\mathbf{y}\|} \geq 0 \\ &\rightarrow \frac{2(\mathbf{x} \cdot \mathbf{y})}{\|\mathbf{x}\|\|\mathbf{y}\|} \geq -2 \\ &\rightarrow \mathbf{x} \cdot \mathbf{y} \geq -\|\mathbf{x}\|\|\mathbf{y}\| \\ &\rightarrow |\mathbf{x} \cdot \mathbf{y}| \leq \|\mathbf{x}\|\|\mathbf{y}\|\end{aligned}$$

Part (c): Applying the suggested method,

$$\begin{aligned}\|\mathbf{x} + \mathbf{y}\|^2 &= (\mathbf{x} + \mathbf{y}) \cdot (\mathbf{x} + \mathbf{y}) \\ &= \mathbf{x} \cdot \mathbf{x} + 2(\mathbf{x} \cdot \mathbf{y}) + \mathbf{y} \cdot \mathbf{y} \\ &= \|\mathbf{x}\|^2 + 2(\mathbf{x} \cdot \mathbf{y}) + \|\mathbf{y}\|^2 \leq \|\mathbf{x}\|^2 + 2\|\mathbf{x}\|\|\mathbf{y}\| + \|\mathbf{y}\|^2 \\ &\implies \|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|\end{aligned}$$

Part (d): In order for d to be a metric, it must be nonnegative, symmetric, and satisfy the triangle inequality. The definition of the Euclidean distance is

$$d(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\| = \sqrt{(x_1 - y_1)^2 + \cdots + (x_n - y_n)^2}$$

Squaring a real number always results in a nonnegative number, and the square root of a nonnegative number is also nonnegative. Thus, $d(\mathbf{x}, \mathbf{y}) \geq 0$. Additionally, $d(\mathbf{x}, \mathbf{y}) = 0$ if and only if $\mathbf{x} = \mathbf{y}$.

Since $|a - b| = |b - a|$, order in the Euclidean distance does not matter. Therefore, d is symmetric.

Finally, we know

$$d(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\| = \|(\mathbf{x} - \mathbf{z}) + (\mathbf{z} - \mathbf{y})\| \leq \|\mathbf{x} - \mathbf{z}\| + \|\mathbf{z} - \mathbf{y}\| = d(\mathbf{x}, \mathbf{z}) + d(\mathbf{z}, \mathbf{y})$$

Thus, d satisfies the triangle inequality, and is a metric.

Problem 6

Define $f_n : [0, 1] \rightarrow \mathbb{R}$ by the equation $f_n(x) = x^n$. Show that the sequence $\{f_n(x)\}$ converges for each $x \in [0, 1]$, but that the sequence $\{f_n\}$ does not converge uniformly. Recall that uniform convergence to f on $[a, b]$ implies that, for any $\varepsilon > 0$, there exists an N such that $|f_n(t) - f(t)| < \varepsilon$ for all $n > N$ and all $t \in [a, b]$.

Solution

For all values of $x \in [0, 1)$, it is apparent that $\lim_{n \rightarrow \infty} x^n = 0$. When $x = 1$, $\lim_{n \rightarrow \infty} 1^n = 1$. This gives

$$f(x) = \lim_{n \rightarrow \infty} f_n(x) = \begin{cases} 0, & x \in [0, 1), \\ 1, & x = 1. \end{cases}$$

As we did in lecture, let's choose some ε that obviously fail the uniform convergence test. Here the gap will be quite large, so we can choose virtually any $\varepsilon < 1$. I'll choose $\varepsilon = \frac{9}{16}$ (the day I'm writing this).

$$|f_n(t) - f(t)| < \varepsilon \implies |t^n| < \frac{9}{16}$$

Choosing $n = N + 1$,

$$t = \left(\frac{9}{16}\right)^{\frac{1}{N+1}}$$

Now

$$|f_{N+1}(t)| = \left(\frac{9}{16}\right)^{\frac{N+1}{N+1}} = \frac{9}{16} \not< \frac{9}{16}$$

Therefore, the sequence does not converge uniformly.